

Part V — Waves, Sound, and Light · Chapter 28

Interference, Diffraction, and Polarization

Wave phenomena that reveal the true nature of light and underlie modern photonics

Hold a CD at an angle under a white light and watch the entire spectrum of colors shift across its surface as you tilt it. The colors come not from pigments but from light waves interfering with each other after reflecting from the microscopic grooves etched in the disc. Blow a soap bubble and watch the iridescent swirls of color that form as the film thins. Press two sheets of glass together and illuminate them with a single-color lamp: you will see concentric rings of brightness and darkness called Newton's rings — mapped by Newton in the 17th century, explained by Young a century later. In each case, the explanation requires treating light as a wave. Ray optics cannot produce interference or diffraction — these are uniquely wave phenomena. This chapter completes the optics unit by revealing the deeper wave nature of light and the phenomena it explains.

28.1 Why Wave Optics Is Needed

The ray model of light (geometric optics) treats light as straight-line rays that reflect and refract at boundaries. It works beautifully for mirrors, lenses, and prisms — systems where wavelength is much smaller than any relevant dimension. But it completely fails to predict some of the most striking behaviors of light.

Geometric optics predicts that a beam of light passing through a narrow slit would produce a sharp-edged bright strip on the far side. In reality, if the slit is narrow enough (comparable to the wavelength), the light fans out broadly beyond the slit, and a complex pattern of bright and dark bands appears. Geometric optics predicts that two beams of light crossing each other simply pass through unchanged. In reality, two beams from the same source crossing at an angle produce regular bands of brightness and darkness: interference fringes.

These phenomena — diffraction and interference — demand the wave model. They are not marginal effects: they occur whenever light interacts with structures on the scale of its wavelength (hundreds of nanometers), which includes almost all fine structures in

nature and technology. The biological optics of butterfly wings and peacock feathers, the spectroscopy used to identify atoms and molecules, the CD and DVD, the anti-reflection coating on camera lenses, the resolution limit of every telescope and microscope, the operation of interferometric gravitational wave detectors — all of these depend on wave optics.

Key Point: A system moves from geometric to wave optics when the wavelength becomes comparable to a relevant dimension. For visible light ($\lambda \approx 500 \text{ nm}$), this means structures smaller than about 0.1 mm begin to require wave treatment.

28.2 Review of Superposition in Waves

Wave optics is built on the superposition principle established in Chapter 24: when two or more waves overlap, the resultant displacement is the algebraic sum of the individual displacements at every point. This is exact for linear media (which includes all optical materials for normal light intensities).

Two key concepts govern the outcome of superposition: phase and coherence. Phase describes where in its oscillation cycle a wave is at a given point. Two waves are in phase if their oscillations align (crest with crest, trough with trough). They are out of phase if one's crest aligns with the other's trough.

Coherence is the property of two waves that enables stable interference. Two waves are coherent if they maintain a fixed phase relationship over time. A laser produces highly coherent light; an incandescent bulb produces incoherent light (random phase relationships between different atomic emissions). Incoherent sources can produce interference, but only over extremely short path-length differences; for the interference patterns studied here, coherent sources (lasers, or light from the same narrow source passed through both slits) are required.

The Rule: For stable interference patterns, the sources must be coherent: they must have the same frequency and a fixed (not random) phase relationship. In practice, this means using the same source for both interfering beams.

28.3 Constructive and Destructive Interference

When two coherent waves of the same frequency and amplitude meet, the nature of their interference depends entirely on the path-length difference δ (delta) between the waves: the difference in distance each wave traveled from its source to the observation point.

Constructive interference occurs when the waves arrive in phase: crest aligns with crest, trough with trough. This happens when the path-length difference is a whole number of wavelengths. The waves reinforce, producing amplitude equal to the sum of the individual amplitudes (and intensity proportional to the square of amplitude).

Destructive interference occurs when the waves arrive exactly out of phase: crest aligns with trough. This happens when the path-length difference is a half-integer number of wavelengths. The waves cancel, producing zero amplitude (and zero intensity for equal-amplitude waves).

$$\text{Constructive: } \delta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots)$$

$$\text{Destructive: } \delta = (m + \tfrac{1}{2})\lambda \quad (m = 0, \pm 1, \pm 2, \dots)$$

Interference conditions based on path-length difference δ . These apply when sources are in phase and both emit the same wavelength.

A phase shift of 180° (half wavelength) can also occur at a reflection boundary when light reflects from a more optically dense medium (higher n). This 'reflection phase shift' changes which path differences give constructive vs. destructive interference in thin-film problems.

28.4 Interference of Light Waves

Optical interference is interference between light waves. Because the wavelength of visible light is roughly 400–700 nm, interference effects are visible only when path differences are controlled on this nanometer scale. Two broad contexts produce visible interference: wavefront division (splitting a single wave front into two, as in a double slit) and amplitude division (splitting the intensity of a single beam into two beams that travel different paths, as in thin films).

Thin-film interference is one of the most common and visually beautiful optical phenomena. When light strikes a thin transparent film (soap bubble, oil slick on water, anti-reflection coating on a lens), it partially reflects from the front surface and partially from the back surface. These two reflected beams travel different path lengths (differing by approximately 2 times the film thickness times the index) and interfere when they reunite. The interference is constructive for some wavelengths and destructive for others, producing the brilliant colors seen in soap films.

Newton's rings are a thin-film interference pattern formed when a slightly curved glass lens rests on a flat glass plate. The air gap between them increases outward from the center, creating a set of concentric circular interference rings. Newton observed these rings in the 17th century, but it was Young in the 19th century who correctly explained

them using wave optics.

Example — Anti-reflection coating A glass lens ($n_{\text{glass}} = 1.52$) is coated with MgF_2 ($n_{\text{coat}} = 1.38$). Thickness = $\lambda/(4n_{\text{coat}})$. For $\lambda = 550$ nm (green): thickness = $550/(4 \times 1.38) \approx 100$ nm. This quarter-wave coating causes destructive interference of reflected green light, greatly reducing glare.

28.5 Young's Double-Slit Experiment

Young's double-slit experiment (1801) provided decisive evidence that light behaves as a wave. Thomas Young illuminated two narrow, closely spaced parallel slits with a single coherent light source. On a screen beyond the slits, instead of the two bright strips predicted by particle/ray optics, he observed a regular pattern of alternating bright and dark bands: interference fringes.

The explanation: each slit acts as a coherent source of circular (cylindrical in 3D) wave fronts. These two sets of wave fronts overlap beyond the slits and interfere. At points on the screen where the path-length difference from the two slits is a whole number of wavelengths, constructive interference produces a bright fringe. Where the path-length difference is a half-integer of wavelengths, destructive interference produces a dark fringe.

For slits separated by distance d , with a screen at distance L (where $L \gg d$), the position y on the screen of the m -th bright fringe (measured from the center) is:

$$y_m = m\lambda L / d \quad (\text{bright fringes, } m = 0, \pm 1, \pm 2, \dots)$$

Bright fringe positions in Young's double-slit experiment. d = slit separation; L = screen distance; λ = wavelength. The central fringe ($m = 0$) is at $y = 0$.

The fringe spacing (distance between adjacent bright fringes) is $\Delta y = \lambda L/d$. Larger wavelength, larger screen distance, or smaller slit separation all increase the fringe spacing. For visible light, fringe spacings of millimeters are easily produced with millimeter-scale slit separations and meter-scale screen distances.

Example — Double-slit fringe spacing Two slits separated by $d = 0.50$ mm = 5.0×10^{-4} m, screen at $L = 2.0$ m, $\lambda = 600$ nm = 6.0×10^{-7} m. Fringe spacing: $\Delta y = \lambda L/d = (6.0 \times 10^{-7} \times 2.0) / (5.0 \times 10^{-4}) = 2.4 \times 10^{-3}$ m = 2.4 mm.

Historical Note: Young's 1801 experiment was met with fierce opposition from supporters of Newton's particle theory of light. Physicists such as Henry Brougham attacked Young's work vigorously. Young responded with characteristic calm precision. It was not until Fresnel's mathematical wave theory (1818) and Foucault's direct

measurement showing light travels slower in water than in air (1850) that the wave theory became universally accepted.

28.6 Diffraction and Wave Spreading

Diffraction is the bending and spreading of a wave as it passes through an opening or around an obstacle. For light, the critical parameter is the ratio λ/a , where a is the width of the opening (or size of the obstacle). When $\lambda \ll a$, almost no diffraction occurs and geometric optics is valid. When $\lambda \approx a$, strong diffraction and complex interference patterns appear.

Diffraction is not an anomaly or edge effect — it is fundamental to wave propagation. Every lens, telescope, and microscope is limited by diffraction. The minimum resolvable detail in any optical instrument is determined not by the quality of the glass but by the diffraction of light at the aperture. This is the diffraction limit (or Rayleigh limit), and it defines the fundamental resolution limit of all wave-based imaging systems.

Rayleigh criterion for resolution: two point objects are just barely resolved when the central diffraction maximum of one falls on the first diffraction minimum of the other. For a circular aperture of diameter D , the minimum angular separation of two resolvable points is:

$$\theta_{\min} \approx 1.22 \lambda / D \quad \text{(Rayleigh criterion for circular aperture)}$$

Diffraction limit for a circular aperture (telescope, eye, camera). Larger aperture D or shorter wavelength λ gives better resolution (smaller $\theta_{\min}$).

The factor 1.22 comes from the mathematics of diffraction from a circular opening. This is why large telescopes are built: larger diameter D gives smaller $\theta_{\min}$, allowing closer objects to be distinguished. The Hubble Space Telescope ($D = 2.4$ m, $\lambda \approx 550$ nm in visible light) has $\theta_{\min} \approx 2.8 \times 10^{-7}$ radians ≈ 0.06 arcseconds — enough to read a newspaper headline from 2 km away.

28.7 Diffraction Through Openings and Around Obstacles

Single-slit diffraction: when light passes through a single slit of width a , the waves from different parts of the slit interfere with each other on the far side. The result is a central bright maximum flanked by alternating dark and bright bands of decreasing intensity. The dark fringes (minima) in single-slit diffraction occur at:

$$a \sin\theta = m\lambda \quad \text{(dark fringes, single slit, } m = \pm 1, \pm 2, \dots)$$

$$\pm 2, \dots)$$

Single-slit diffraction minima. a = slit width; θ = angle from central axis; m = integer $\neq 0$. Note: $m = 0$ is the central bright maximum, not a minimum.

The central maximum has angular half-width $\theta \approx \lambda/a$. Narrow slits (small a) produce wide diffraction patterns; wide slits produce narrow patterns. This inverse relationship is a fundamental feature of diffraction: concentrating a wave in one dimension (narrow slit) causes it to spread in the perpendicular dimension.

Babinet's Principle: the diffraction pattern from an opaque obstacle has the same intensity distribution as the pattern from an aperture of the same shape and size (except at the exact forward direction). This is why the shadow of a circular disk has a bright spot at its very center (Poisson's spot or Arago spot) — the diffraction from the circular edge constructively interferes on-axis. This was experimentally predicted as a consequence of Fresnel's wave theory and confirmed by Arago in 1818, converting many remaining wave-theory skeptics.

28.8 Diffraction Gratings

A diffraction grating is an optical element with a large number of equally spaced, parallel slits or grooves (typically thousands per millimeter). Each slit acts as a coherent source; the combined interference of waves from all slits produces extremely sharp, bright maxima at specific angles.

The grating equation gives the angles at which constructive interference produces bright maxima:

$$d \sin\theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \dots)$$

Grating equation. d = grating spacing (distance between adjacent slits); θ_m = angle of m -th order maximum; λ = wavelength.

Different wavelengths satisfy the grating equation at different angles, so a grating disperses white light into a complete spectrum much more cleanly and brightly than a prism. The zeroth order ($m = 0$) is always at $\theta = 0$ regardless of wavelength. The first-order ($m = 1$) maxima appear at different angles for each wavelength. Higher orders appear at larger angles.

Diffraction gratings are the key element in spectroscopes and spectrometers — instruments that measure the wavelengths of light emitted or absorbed by atoms and molecules. Every element has a unique emission spectrum (set of specific wavelengths), like a fingerprint. Grating spectroscopy allows astronomers to determine the composition, temperature, and velocity of stars and galaxies from their light; allows chemists to identify compounds; and allows physicists to verify quantum mechanical

predictions about atomic energy levels.

Example — Grating calculation A grating has 600 lines/mm, so $d = 1/600 \text{ mm} = 1.67 \times 10^{-6} \text{ m}$. For $\lambda = 550 \text{ nm}$, first-order maximum: $\sin\theta_1 = \lambda/d = 5.50 \times 10^{-7} / 1.67 \times 10^{-6} = 0.329$. $\theta_1 = \arcsin(0.329) = 19.2^\circ$.

28.9 Polarization of Light

Polarization is a property of transverse waves in which the direction of oscillation is restricted. For a transverse wave, the oscillation can occur in any direction perpendicular to propagation. Unpolarized light (like sunlight or incandescent light) has its electric field vectors oscillating in all perpendicular directions with equal probability and random phase. Polarized light has its electric field oscillating preferentially in one direction.

Linear polarization: the electric field oscillates along a single direction (the polarization axis). Circular polarization: the electric field rotates uniformly around the propagation direction as the wave travels. Elliptical polarization is the general case, of which linear and circular are special cases.

Three mechanisms produce polarized light: selective absorption (polarizing filters), reflection at Brewster's angle, and scattering. In everyday life, reflected light from non-metallic surfaces (water, glass, road, snow) is partially polarized parallel to the surface. Polarizing sunglasses exploit this by transmitting only vertically polarized light, blocking the horizontally polarized reflected glare.

28.10 Polarizing Filters

A polarizing filter (polarizer) transmits only the component of the electric field along its transmission axis and absorbs the perpendicular component. When unpolarized light of intensity I_0 passes through an ideal polarizer, the transmitted intensity is exactly $I_0/2$, because the filter passes only one of the two equally probable oscillation directions.

When polarized light of intensity I passes through a second polarizer (called an analyzer) whose transmission axis makes angle θ with the polarization direction of the incoming light, the transmitted intensity is given by Malus's Law:

$$I = I_0 \cos^2 \theta \quad (\text{Malus's Law})$$

Malus's Law. I_0 = intensity of incident polarized light; θ = angle between polarization direction and analyzer axis. Maximum at $\theta = 0^\circ$; zero at $\theta = 90^\circ$ (crossed polarizers).

When two polarizers are crossed (transmission axes at 90°), no light passes through:

$\cos^2(90^\circ) = 0$. This is easily demonstrated: two polarizing filters held at right angles produce a completely dark field. Inserting a third polarizer between them at 45° to each will restore some light, which appears counterintuitive but follows directly from Malus's Law applied sequentially to each polarizer.

Example — Malus's Law Polarized light at $I_0 = 20 \text{ W/m}^2$ passes through an analyzer at $\theta = 35^\circ$. $I = 20 \times \cos^2(35^\circ) = 20 \times (0.8192)^2 = 20 \times 0.671 = 13.4 \text{ W/m}^2$.

Liquid crystal displays (LCDs) work by sandwiching a liquid crystal layer between two crossed polarizers. The liquid crystal can rotate the polarization of light by 90° when no voltage is applied (letting light through) or leave it unchanged when voltage is applied (blocking light). Switching individual pixels on and off with voltage produces the images on every LCD screen.

28.11 Evidence That Light Is Transverse

The existence of polarization is definitive proof that light is a transverse wave, not a longitudinal one. A longitudinal wave — like sound — oscillates parallel to its propagation direction; there is only one possible oscillation direction, and polarization is impossible. A transverse wave oscillates perpendicular to propagation, allowing oscillations in many directions — and thus allowing polarization.

Sound cannot be polarized; light can. This experimental fact establishes that light must be transverse. The direction of the electric field's oscillation is the polarization direction of light. Electromagnetic wave theory confirms this: the electric and magnetic fields in a light wave are always perpendicular to the direction of propagation and to each other.

Brewster's angle (also called the polarizing angle) provides another demonstration. When light strikes a dielectric surface at Brewster's angle $\theta_B = \arctan(n_2/n_1)$, the reflected light is completely polarized parallel to the surface (s-polarized). The refracted light is partially polarized. Brewster discovered this in 1815; it is explained by electromagnetic wave theory and is only possible because light is transverse.

$$\theta_B = \arctan(n_2 / n_1) \quad (\text{Brewster's angle})$$

Brewster's angle: the angle of incidence at which the reflected light is completely linearly polarized. At this angle, the reflected and refracted rays are perpendicular to each other.

28.12 Real-World Applications of Wave Optics

Wave optics underpins an enormous range of practical technologies, many of which would be impossible to engineer without understanding interference, diffraction, and

polarization.

Anti-reflection coatings

Camera lenses, microscope objectives, and eyeglasses are coated with thin dielectric films (typically MgF_2 or multiple layers of other materials) designed so that reflected light undergoes destructive interference. A single quarter-wave coating can reduce reflection from 4% to less than 0.5% for a specific wavelength. Multi-layer coatings (10–50 layers) can reduce reflection to below 0.1% across the entire visible spectrum.

Holography

A hologram records the interference pattern between a reference beam and a beam scattered from an object. When the hologram is illuminated by a laser, it reconstructs the wavefronts from the object, producing a three-dimensional image. Credit card security holograms, medical imaging, and data storage systems all use holographic principles.

Interferometers

Interferometers split a beam into two paths, send them along different routes, and recombine them to measure path-length differences with sub-wavelength precision. The Michelson interferometer (1887) was used to show that the speed of light is the same in all directions, contributing to the development of special relativity. The LIGO (Laser Interferometer Gravitational-Wave Observatory) uses two 4 km arms; when a gravitational wave passes, one arm stretches by 10^{-18} m — less than one-ten-thousandth the diameter of a proton — which is detected as a change in the interference pattern.

Optical data storage (CD, DVD, Blu-ray)

CDs use pits and lands ($0.83\text{ }\mu\text{m}$ wide) etched in a reflective aluminum layer. A laser reads the disc: when the beam crosses a pit edge, reflected light from the pit and the surrounding land interfere destructively, signaling a bit transition. DVD uses smaller features ($0.74\text{ }\mu\text{m}$) and a shorter-wavelength laser for higher density. Blu-ray uses a 405 nm blue laser for even finer features ($0.32\text{ }\mu\text{m}$) and higher capacity.

X-ray crystallography

X-rays have wavelengths ($0.01\text{--}1\text{ nm}$) comparable to interatomic spacings in crystals. When X-rays diffract from a crystal's periodic lattice, they produce a diffraction pattern from which the crystal structure can be determined with atomic resolution. DNA, proteins, and drugs have all had their molecular structures determined using X-ray crystallography.

"No experiment has done more to reveal the wave nature of matter and radiation than diffraction. Every structure we have determined — from salt crystals to the double helix — relied on wave interference at the atomic scale." — Condensed from the history of crystallography

28.13 Common Mistakes in Wave Optics Analysis

- Applying the double-slit constructive interference formula $\delta = m\lambda$ to single-slit diffraction. Single-slit minima occur at a $\sin\theta = m\lambda$; single-slit maxima do not occur at simple integer wavelengths. These are different formulas for different geometries.
- Confusing the grating equation with the double-slit formula. The grating equation $d \sin\theta = m\lambda$ and the double-slit condition $\delta = m\lambda$ (for small angles, where $\delta = d \sin\theta$) are identical in form, but a grating has thousands of slits while the double slit has two. The grating produces much sharper, brighter maxima.
- Forgetting phase shifts on reflection in thin-film problems. Light reflecting from a medium of higher index undergoes a 180° phase shift (like a string reflecting from a fixed end). If one reflected beam has a phase shift and the other does not, the constructive and destructive conditions are reversed. Always check each interface.
- Misidentifying the order of the minimum in single-slit diffraction. The central bright maximum is at $\theta = 0$ ($m = 0$). The first minimum is at a $\sin\theta = \pm\lambda$ ($m = \pm 1$). The second bright maximum (approximately) is between $m = 1$ and $m = 2$ minima. Students often call the first bright side maximum 'm = 1' and confuse it with the $m = 1$ minimum.
- Applying Malus's Law to unpolarized light. Malus's Law $I = I_0 \cos^2\theta$ applies only when the incident light is already linearly polarized. For unpolarized light passing through a single polarizer, the transmitted intensity is $I_0/2$, regardless of the polarizer orientation.
- Treating the Rayleigh criterion as an absolute limit. The Rayleigh criterion $\theta_{\min} = 1.22 \lambda/D$ gives the conventional resolution limit for a circular aperture, but it is a criterion (a convention), not a physical law. Super-resolution techniques can exceed this limit using structured illumination, near-field optics, or single-molecule localization.

Common Mistake: In wave optics, always identify wavelength in the medium ($\lambda_n = \lambda_0/n$) when calculating path differences inside a material. The path-length difference

in the material must be compared to the wavelength in that material, not the vacuum wavelength.

Bridge Note *In quantum mechanics, the wave-particle duality extends interference and diffraction to matter. Electrons, neutrons, and even large molecules undergo double-slit interference with de Broglie wavelength $\lambda = h/p$. The mathematical description — superposition of quantum amplitudes, Born rule for probabilities — is a direct generalization of classical wave superposition. The concepts introduced in this chapter are the exact foundation for quantum wave mechanics.*

Chapter Summary

Wave optics is required when wavelength is comparable to relevant dimensions. Ray optics cannot explain interference, diffraction, or polarization.

- Superposition of coherent waves produces constructive interference ($\delta = m\lambda$) and destructive interference ($\delta = (m + \frac{1}{2})\lambda$).
- Young's double slit produces bright fringes at $y_m = m\lambda L/d$ and fringe spacing $\Delta y = \lambda L/d$. It established light as a wave.
- Single-slit diffraction produces a central maximum with minima at $a \sin\theta = m\lambda$ ($m \neq 0$). Narrower slit $\rightarrow$ wider pattern.
- Diffraction limit (Rayleigh): $\theta_{\min} = 1.22\lambda/D$. Larger aperture or shorter wavelength gives better resolution.
- Diffraction grating: $d \sin\theta = m\lambda$. Produces sharp, bright maxima; separates wavelengths for spectroscopy.
- Polarization proves light is transverse. Unpolarized light through a polarizer gives $I/2$. Malus's Law: $I = I_0 \cos^2\theta$ for incident polarized light.
- Brewster's angle: $\theta_B = \arctan(n_2/n_1)$. Reflected light is fully polarized at this angle.
- Applications: anti-reflection coatings, holography, interferometers, optical storage, X-ray crystallography, LCD screens, polarizing sunglasses.

Key Terms

wave optics (physical optics): The treatment of light as a wave, necessary when wavelength is comparable to relevant dimensions; explains interference, diffraction,

and polarization.

coherence: The property of two wave sources that maintain a fixed phase relationship over time; required for stable interference patterns.

path-length difference (δ): The difference in distances traveled by two interfering waves from their sources to the observation point; determines whether interference is constructive or destructive.

constructive interference: Interference in which waves arrive in phase ($\delta = m\lambda$); amplitudes add; bright fringe in optical interference.

destructive interference: Interference in which waves arrive out of phase ($\delta = (m + \frac{1}{2})\lambda$); amplitudes cancel; dark fringe in optical interference.

thin-film interference: Interference of light reflected from the front and back surfaces of a thin transparent film; produces colored patterns in soap bubbles, oil slicks, and coatings.

reflection phase shift: A 180° phase change that occurs when light reflects from a medium of higher index of refraction; equivalent to a half-wavelength path difference.

Young's double-slit experiment: The landmark 1801 experiment by Thomas Young demonstrating that light produces interference fringes, establishing its wave nature.

interference fringe: A bright or dark band in an interference pattern; bright fringes are constructive maxima, dark fringes are destructive minima.

fringe spacing (Δy): The distance between adjacent bright (or dark) fringes in a double-slit pattern: $\Delta y = \lambda L/d$.

diffraction: The bending and spreading of a wave as it passes through an aperture or around an obstacle; most pronounced when $\lambda \approx$ aperture size.

diffraction limit (Rayleigh criterion): The minimum angular separation of two resolvable point objects: $\theta_{\min} = 1.22\lambda/D$ for a circular aperture of diameter D .

single-slit diffraction: The interference of waves from different parts of a single slit, producing a central maximum and side minima at a $\sin\theta = m\lambda$.

diffraction grating: An optical element with many equally spaced slits or grooves; produces sharp spectral maxima obeying $d \sin\theta = m\lambda$.

polarization: The restriction of a transverse wave's oscillation to a preferred direction; only possible for transverse waves (not longitudinal).

polarizing filter (polarizer): An optical element that transmits only the component of the electric field along its transmission axis.

Malus's Law: $I = I_0 \cos^2\theta$; the intensity transmitted through an analyzer after polarized light passes through it at angle θ to the polarization axis.

Brewster's angle (θ_B): $\theta_B = \arctan(n_2/n_1)$; the angle of incidence at which reflected

light is completely linearly polarized parallel to the surface.

interferometer: An optical instrument that splits and recombines a beam of light to measure path-length differences with sub-wavelength precision.

hologram: A recording of the interference pattern between a reference beam and a scattered beam; reconstructs a three-dimensional image when illuminated.

Concept Review Questions

Answer in complete sentences. No calculations required unless specified.

1. Explain why geometric (ray) optics fails to account for interference and diffraction. Give a specific example where the ray model makes a prediction that contradicts experiment.
 2. Describe Young's double-slit experiment. What pattern did Young observe, and why is it impossible to explain with the particle model of light?
 3. Why does a thin soap film appear colored in white light? Would the film appear the same color from all viewing angles? Explain why or why not.
 4. Explain why the central fringe in a double-slit pattern is bright. A student says: "The two paths are equal, so they cancel." What is wrong with this reasoning?
 5. Why do astronomers build telescopes with large mirrors or lenses? Use the diffraction limit formula to explain why doubling the aperture improves resolution.
 6. Why is polarization evidence that light is transverse? Explain why sound cannot be polarized.
 7. Two polarizing filters are crossed (90° apart) and no light passes through. A third filter is inserted between them at 45° to each. Explain why some light now passes through, using Malus's Law.
 8. Explain how a diffraction grating works as a spectrometer. Why can a grating separate colors more precisely than a prism?
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Interference Problems

9. Two slits are separated by 0.30 mm. Red light ($\lambda = 660$ nm) illuminates the slits and a screen is placed 1.5 m away. Find (a) the fringe spacing, (b) the position of the 3rd bright fringe, (c) the position of the 2nd dark fringe.
 10. A double-slit experiment uses green light ($\lambda = 532$ nm). The fringe spacing is 3.2 mm and the screen is 1.8 m from the slits. Find the slit separation.
 11. A thin film of oil ($n = 1.45$) 200 nm thick lies on water ($n = 1.33$). White light reflects from the film. Which visible wavelengths (400–700 nm) undergo constructive interference in the reflected light? (Note: one phase shift on reflection from oil surface, none from oil–water since $n_{\text{oil}} > n_{\text{water}}$.)
 12. An anti-reflection coating of MgF_2 ($n = 1.38$) is applied to glass ($n = 1.52$). For the coating to cause destructive interference of reflected light at $\lambda = 550$ nm, what minimum coating thickness is required? (Both reflections have phase shifts: air– MgF_2 and MgF_2 –glass.)
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Diffraction Exercises

13. A single slit of width 0.40 mm is illuminated by $\lambda = 600$ nm light at a screen 1.50 m away. Find (a) the angular half-width of the central maximum, (b) the width of the central maximum on the screen.
 14. A diffraction grating has 500 lines/mm. Find the angle of the first-order maximum for (a) red light ($\lambda = 700$ nm) and (b) violet light ($\lambda = 400$ nm).
 15. A telescope objective has diameter $D = 150$ mm. Find the angular resolution limit (Rayleigh criterion) for $\lambda = 550$ nm. Convert to arcseconds. Can this telescope resolve two stars separated by 0.8 arcseconds?
 16. X-rays with $\lambda = 0.154$ nm (characteristic copper X-ray) diffract from a crystal with interplanar spacing $d = 0.200$ nm. Using the Bragg condition $2d \sin\theta = m\lambda$, find the angle θ for $m = 1$. (Bragg diffraction is the grating equation applied to reflection from crystal planes.)
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Polarization Analysis

17. Unpolarized light of intensity 80 W/m^2 passes through a polarizer. What intensity emerges?

18. Light polarized vertically at intensity 60 W/m^2 passes through an analyzer with its axis at (a) 0° , (b) 30° , (c) 60° , (d) 90° from vertical. Find the transmitted intensity in each case.
 19. Horizontally polarized light passes through three consecutive polarizers: the first at 30° from horizontal, the second at 60° , the third at 90° (vertical). If $I_0 = 100 \text{ W/m}^2$, find the intensity after each polarizer.
 20. Find Brewster's angle for light going from air ($n = 1.00$) into glass ($n = 1.55$). At this angle, what is the angle of refraction in the glass? (Use Snell's Law to verify that the reflected and refracted rays are perpendicular: $\theta_B + \theta_{\text{refracted}} = 90^\circ$.)
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Applied Wave Optics Problems

21. A CD has a track spacing of $1.6 \text{ }\mu\text{m}$. It is read with a $\lambda = 780 \text{ nm}$ laser. What angle does the first-order diffracted beam make with the normal to the disc surface?
22. The LIGO gravitational wave detector uses $\lambda = 1064 \text{ nm}$ infrared laser light in interferometer arms 4.0 km long. A gravitational wave stretches one arm by $\Delta L = 10^{-18} \text{ m}$. Express this as a fraction of the wavelength. What is the corresponding change in path-length difference between the two arms?
23. A butterfly's wing produces iridescent blue color (peak $\lambda \approx 450 \text{ nm}$) through thin-film interference in chitin nanostructures ($n = 1.56$). Estimate the film thickness that produces this color for near-normal reflection. (Assume one phase-shift reflection; constructive condition: $2nt = \lambda$, so $t = \lambda/(2n)$.)
24. An LCD pixel is formed by two crossed polarizers with a liquid crystal layer between them. In the 'on' state, the crystal rotates polarization by 90° , allowing light through. In the 'off' state, no rotation occurs. Using Malus's Law, explain why the pixel appears bright in the on state and dark in the off state. What percentage of the original unpolarized light reaches the viewer in the bright state (accounting for both polarizers)?

Chapter 28 Checkpoint

Before moving on, confirm you can do each of the following.

I can...	See Section
Explain why wave optics is needed and give an example where ray optics fails	28.1
Define coherence and explain why it is required for stable interference	28.2
State the conditions for constructive and destructive interference using path difference	28.3
Explain thin-film interference and the role of phase shifts on reflection	28.4
Describe Young's double-slit experiment and calculate fringe positions and spacing	28.5
Explain diffraction and the Rayleigh criterion for resolution	28.6
Find single-slit diffraction minima using a $\sin\theta = m\lambda$	28.7
Apply the grating equation $d \sin\theta = m\lambda$ and calculate diffraction angles	28.8
Define polarization and explain why it is evidence that light is transverse	28.9
Apply Malus's Law to calculate transmitted intensity through an analyzer	28.10
Calculate Brewster's angle and state what happens at that angle	28.11
Describe three technological applications of wave optics	28.12
Identify and avoid common errors in wave optics calculations	28.13

End of Part V: Waves, Sound, and Light — Continue to Part VI: Electricity and Magnetism →